Министерство образования Республики Беларусь
Учреждение образования
БЕЛОРУССКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ
ИНФОРМАТИКИ И РАДИОЭЛЕКТРОНИКИ

Факультет компьютерных систем и сетей
Кафедра информатики
Дисциплина: Избранные главы информатики

ОТЧЁТ

к лабораторной работе №2
на тему:

«Система контейнеризации Docker»

Выполнил: студент группы 453502
Шавловский Станислав Андреевич
Проверила: Жвакина Анна Васильевна

Минск 2026

Цель: Познакомиться с возможностями и получить практические навыки работы с Docker.

Задание:

1. *Подготовка рабочего окружения*

   a. Установка Docker

```
PS C:\Users\User> docker --version
Docker version 29.2.1, build a5c7197
PS C:\Users\User> docker compose version
Docker Compose version v5.0.2
```

   b. Выполнение базовых настроек

```
PS C:\Users\User> docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.
```

```
PS C:\Users\User> docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
74c3f955a1d5   bridge    bridge    local
eabc8a17436e   host      host      local
7905d7d158fa   none      null      local
```

```
PS C:\Users\User> docker images
                                                                                  i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   ef54e839ef54       25.9kB         9.52kB    U
```

```
PS C:\Users\User> docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS                     PORTS     NAMES
321ec8e74795   hello-world   "/hello"   31 seconds ago   Exited (0) 30 seconds ago            eager_mccarthy
```

2. *Изучите простейшие консольные команды и возможности Docker Desktop (см. лекцию), создать собственный контейнер docker/getting-started, открыть в браузере и изучить tutorial*

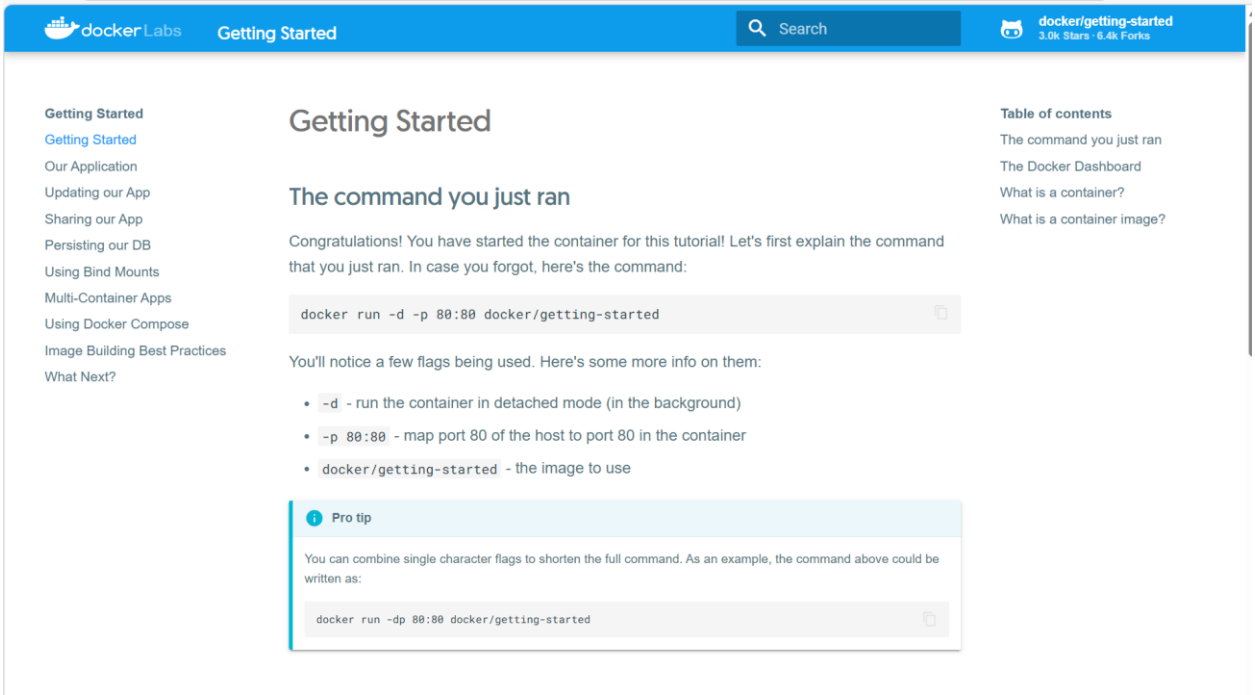
```
PS C:\Users\User> docker pull docker/getting-started
Using default tag: latest
latest: Pulling from docker/getting-started
4f7e34c2de10: Pull complete
f1d1c9928c82: Pull complete
b6334b6ace34: Pull complete
9b6f639ec6ea: Pull complete
c158987b0551: Pull complete
1e35f6679fab: Pull complete
ee68d3549ec8: Pull complete
cb9626c74200: Pull complete
33e0cbbb4673: Pull complete
Digest: sha256:d79336f4812b6547a53e735480dde67f8f8f7071b414fbd9297609ffb989abc1
Status: Downloaded newer image for docker/getting-started:latest
docker.io/docker/getting-started:latest
```

```
PS C:\Users\User> docker run -d -p 80:80 --name getting-started docker/getting-started
7c78acb56ef5d1c57109478166b74dc386c620e42ac552bec39996ae5aeb93c1
```

```
PS C:\Users\User> docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS
                        NAMES
7c78acb56ef5   docker/getting-started   "/docker-entrypoint.…"   About a minute ago   Up About a minute
/tcp, [::]:80->80/tcp   getting-started
```

```
PS C:\Users\User> docker ps -a
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS
                           NAMES
7c78acb56ef5   docker/getting-started   "/docker-entrypoint.…"   4 minutes ago    Up 4 minutes
->80/tcp, [::]:80->80/tcp   getting-started
321ec8e74795   hello-world              "/hello"                 18 minutes ago   Exited (0) 18 minutes ag
                           eager_mccarthy
```

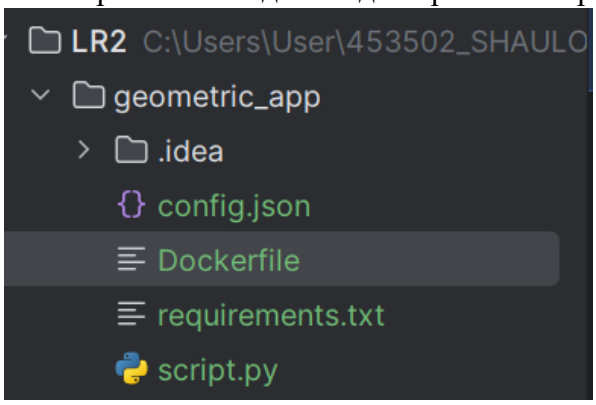
```
PS C:\Users\User> docker logs getting-started
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/10 17:13:41 [notice] 1#1: using the "epoll" event method
2026/03/10 17:13:41 [notice] 1#1: nginx/1.23.3
2026/03/10 17:13:41 [notice] 1#1: built by gcc 12.2.1 20220924 (Alpine 12.2.1_git20220924-r4)
2026/03/10 17:13:41 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/10 17:13:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/10 17:13:41 [notice] 1#1: start worker processes
2026/03/10 17:13:41 [notice] 1#1: start worker process 30
2026/03/10 17:13:41 [notice] 1#1: start worker process 31
2026/03/10 17:13:41 [notice] 1#1: start worker process 32
2026/03/10 17:13:41 [notice] 1#1: start worker process 33
2026/03/10 17:13:41 [notice] 1#1: start worker process 34
2026/03/10 17:13:41 [notice] 1#1: start worker process 35
2026/03/10 17:13:41 [notice] 1#1: start worker process 36
2026/03/10 17:13:41 [notice] 1#1: start worker process 37
2026/03/10 17:13:41 [notice] 1#1: start worker process 38
2026/03/10 17:13:41 [notice] 1#1: start worker process 39
2026/03/10 17:13:41 [notice] 1#1: start worker process 40
```




```
PS C:\Users\User> docker stop getting-started
getting-started
```

3. Создайте docker image, который запускает скрипт с использованием функций из https://github.com/smartiqaorg/geometric_lib.
   a. Данные необходимые для работы скрипта передайте любым удобным способом (например: конфиг файл через docker volume, переменные окружения, перенаправление ввода). Изучите простейшие консольные

команды для работы с docker(см. лекцию). Зарегистрируйтесь на DockerHub и выберите необходимые для проекта образы



b. Создать Dockerfile для реализации сборки собственных Docker образов

```
FROM python:3.10-slim

WORKDIR /app

RUN apt-get update && apt-get install -y git

RUN git clone https://github.com/smartiqaorg/geometric_lib.git

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt || true

COPY . .

ENV PYTHONPATH="/app/geometric_lib:${PYTHONPATH}"

CMD ["python", "script.py"]
```

c. Использовать его для создания контейнера. Протестировать использование контейнера

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2> docker build -t stanislav/geometric-app ./geometric_app
[+] Building 22.9s (13/13) FINISHED                                                                    docker:de
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 383B
 => [internal] load metadata for docker.io/library/python:3.10-slim
 => [auth] library/python:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/7] FROM docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab2
 => => resolve docker.io/library/python:3.10-slim@sha256:4b0a8ebf16cf4563f3d3732bd4f4a464abb2f671b3b9d00aab2
 => [internal] load build context
 => => transferring context: 839B
 => CACHED [2/7] WORKDIR /app
 => CACHED [3/7] RUN apt-get update && apt-get install -y git
 => [4/7] RUN git clone https://github.com/smartiqaorg/geometric_lib.git
 => [5/7] COPY requirements.txt .
 => [6/7] RUN pip install --no-cache-dir -r requirements.txt || true
 => [7/7] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:fa4848819bf17a908d91ddb722e845c9fe352780dbb387325e6a92834f7168c8
 => => exporting config sha256:822a9e601d85f9c3169231f8a701e955a693075dffe603262177f5e47968875f
 => => exporting attestation manifest sha256:e22a7c58015adc585b54efae8a507cec4fcaf119e3d7f491bdb8541d15f6d5b
 => => exporting manifest list sha256:04f33b57c2dee817b891fc2061e106cf99d7b310e8a97fa0cc982436b5028691
 => => naming to docker.io/stanislav/geometric-app:latest
 => => unpacking to docker.io/stanislav/geometric-app:latest
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2> docker run --rm -v "${PWD}/geometric_app/config.json:/app/config.json" stanislav/geometric-app
 Геометрические вычисления
Площадь квадрата со стороной 7: 49
Площадь круга радиуса 10: 314.1592653589793
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2> docker run  -v "${PWD}/geometric_app/config.json:/app/config.json" stanislav/geometric-app
 Геометрические вычисления
Площадь квадрата со стороной 7: 49
Площадь круга радиуса 10: 314.1592653589793
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2> docker ps -a
CONTAINER ID   IMAGE                    COMMAND                  CREATED              STATUS                          PORTS     NAMES
507ad1947443   stanislav/geometric-app   "python script.py"       About a minute ago   Exited (0) About a minute ago             distracted_vi
araya
7c78acb56ef5   docker/getting-started    "/docker-entrypoint.…"   About an hour ago    Exited (0) About an hour ago              getting-start
321ec8e74795   hello-world               "/hello"                 2 hours ago          Exited (0) 2 hours ago                    eager_mccarth
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2> docker start distracted_visvesvaraya
distracted_visvesvaraya
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2> docker logs distracted_visvesvaraya
 Геометрические вычисления
Площадь квадрата со стороной 7: 49
Площадь круга радиуса 10: 314.1592653589793
 Геометрические вычисления
Площадь квадрата со стороной 7: 49
Площадь круга радиуса 10: 314.1592653589793
```

4. Скачать любой доступный проект с GitHub с произвольным стеком технологий (пример – см. индивидуальное задание) или использовать свой, ранее разработанный. Создать для него необходимый контейнер, используя Docker Compose для управления многоконтейнерными приложениями. Запустить проект в контейнере.(Примеры Images: https://hub.docker.com/_/phpmyadmin, https://hub.docker.com/_/mysql, https://hub.docker.com/_/postgres)

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2> git clone https://github.com/RameshMF/springboot-thymeleaf-crud-pagination-sorting-webapp.git springboot_crud_app
Cloning into 'springboot_crud_app'...
remote: Enumerating objects: 58, done.
remote: Counting objects: 100% (20/20), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 58 (delta 9), reused 4 (delta 4), pack-reused 38 (from 1)
Receiving objects: 100% (58/58), 16.63 KiB | 774.00 KiB/s, done.
Resolving deltas: 100% (10/10), done.
```

README.md | Dockerfile | docker-compose.yml

```dockerfile
FROM maven:3.9.9-eclipse-temurin-17 AS build
WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn -q -DskipTests package

FROM eclipse-temurin:17-jdk-jammy
WORKDIR /app
COPY --from=build /app/target/*.jar app.jar

ENV SPRING_DATASOURCE_URL=jdbc:mysql://db:3306/employee_management_system
ENV SPRING_DATASOURCE_USERNAME=root
ENV SPRING_DATASOURCE_PASSWORD=root
ENV SPRING_JPA_HIBERNATE_DDL_AUTO=update

EXPOSE 8080
ENTRYPOINT ["java","-jar","app.jar"]
```

```yaml
version: "3.9"

services:
  db:
    image: mysql:8.0
    container_name: employee-mysql
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: employee_management_system
    ports:
      - "3307:3306"
    volumes:
      - db_data:/var/lib/mysql
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost"]
      interval: 5s
      timeout: 3s
      retries: 10

  app:
    build: .
    container_name: employee-app
    depends_on:
      db:
        condition: service_healthy
    ports:
      - "8080:8080"
    environment:
      SPRING_DATASOURCE_URL: jdbc:mysql://db:3306/employee_management_system
      SPRING_DATASOURCE_USERNAME: root
      SPRING_DATASOURCE_PASSWORD: root
      SPRING_JPA_HIBERNATE_DDL_AUTO: update

volumes:
  db_data:
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker compose build
time="2026-03-11T14:09:18+03:00" level=warning msg="C:\\Users\\User\\453502_SHAULOUSKI_
26\\IGI\\LR2\\springboot_crud_app\\docker-compose.yml: the attribute `version` is obsol
ete, it will be ignored, please remove it to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 601B 0.1s done
#1 DONE 0.1s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 505B 0.0s done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/maven:3.9.9-eclipse-temurin-17
#3 ...

#4 [auth] library/maven:pull token for registry-1.docker.io
#4 DONE 0.0s

#5 [auth] library/eclipse-temurin:pull token for registry-1.docker.io
#5 DONE 0.0s

#6 [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-jammy
#6 DONE 2.2s

#3 [internal] load metadata for docker.io/library/maven:3.9.9-eclipse-temurin-17
#3 DONE 2.3s

#7 [internal] load .dockerignore
#7 transferring context: 2B done
#7 DONE 0.1s

#8 [internal] load build context
#8 DONE 0.0s

#9 [stage-1 1/3] FROM docker.io/library/eclipse-temurin:17-jdk-jammy@sha256:ce06e86f8d5
adea5e0e0f68c63f322dc1cbbbcd3c2875ff8241b39a57083b164
#9 resolve docker.io/library/eclipse-temurin:17-jdk-jammy@sha256:ce06e86f8d5adea5e0e0f6
8c63f322dc1cbbbcd3c2875ff8241b39a57083b164 0.1s done
#9 DONE 0.1s

#10 [build 1/5] FROM docker.io/library/maven:3.9.9-eclipse-temurin-17@sha256:f58d59b627
3e785ac0a4477f6e9b5ba1d7731c75b906c0f7b34076f1851318cc
#10 resolve docker.io/library/maven:3.9.9-eclipse-temurin-17@sha256:f58d59b6273e785ac0a
4477f6e9b5ba1d7731c75b906c0f7b34076f1851318cc
#10 resolve docker.io/library/maven:3.9.9-eclipse-temurin-17@sha256:f58d59b6273e785ac0a
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker images

                                                    1 Info →   U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/getting-started:latest        d79336f4812b       73.9MB         21.4MB    U
hello-world:latest                   ef54e839ef54       25.9kB         9.52kB    U
springboot_crud_app-app:latest       c8cda589b7f9        713MB          235MB
stanislav/geometric-app:latest       2bb462c68996        387MB          103MB    U
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker compose up
time="2026-03-11T14:21:51+03:00" level=warning msg="C:\\Users\\User\\453502_SHAULOUSKI_
26\\IGI\\LR2\\springboot_crud_app\\docker-compose.yml: the attribute `version` is obsol
ete, it will be ignored, please remove it to avoid potential confusion"
[+] up 14/14
 ✔Image mysql:8.0 Pulled
                                                                    82.3s
#1 [internal] load local bake definitions
#1 reading from stdin 601B 0.1s done
#1 DONE 0.1s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 505B 0.0s done
#2 DONE 0.0s

#3 [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk-jammy
#3 ...

#4 [internal] load metadata for docker.io/library/maven:3.9.9-eclipse-temurin-17
#4 DONE 0.6s
```

```
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/getting-started:latest        d79336f4812b       73.9MB         21.4MB    U
hello-world:latest                   ef54e839ef54       25.9kB         9.52kB    U
mysql:8.0                            a3dff78d8762       1.08GB          247MB    U
springboot_crud_app-app:latest       b82bd7bd3662        713MB          235MB    U
stanislav/geometric-app:latest       2bb462c68996        387MB          103MB    U
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app>
```

springboot_cruc	-	-	-		N/A	9 mi	
employee-my	6b4f739f88ad	mysql:8.0	3307:3306		N/A	9 mi	
employee-ap	92263c61aacf	springboot	8080:8080		N/A	9 mi	

Employees List

Add Employee

Employee First Name	Employee Last Name	Employee Email	Actions
rrrr	stanislau	Shavlovsijstanislav3211@mail.ru	Update Delete

5. Настроить сети и тома для обеспечения связи между контейнерами и сохранения данных (исходные данные, логин, пароль и т.д.)

```yaml
services:
  db:
    image: mysql:8.0
    container_name: employee-mysql
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: employee_management_system
    ports:
      - "3307:3306"
    volumes:
      - db_data:/var/lib/mysql
    networks:
      - app_net
    healthcheck:
      test: ["CMD", "mysqladmin", "ping", "-h", "localhost"]
      interval: 5s
      timeout: 3s
      retries: 10

  app:
    build: .
    container_name: employee-app
    depends_on:
      db:
        condition: service_healthy
    ports:
      - "8080:8080"
    environment:
      SPRING_DATASOURCE_URL: jdbc:mysql://db:3306/employee_management_system
      SPRING_DATASOURCE_USERNAME: root
      SPRING_DATASOURCE_PASSWORD: root
      SPRING_JPA_HIBERNATE_DDL_AUTO: update
    volumes:
      - app_logs:/app/logs
    networks:
      - app_net

volumes:
  db_data:
  app_logs:
```

```yaml
networks:
  app_net:
    driver: bridge
```

Volumes Give feedback

Search

Name	Created	Size	Actions
springboot_crud_app_app_logs	5 minutes ago	0 Bytes	
springboot_crud_app_db_data	35 minutes ago	0 Bytes	

6. Разместите результат в созданный репозиторий в DockerHub

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker login
Authenticating with existing credentials... [Username: stasname]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker tag springboot_crud_app-app stasname/springboot-crud-app:latest
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker push stasname/springboot-crud-app:latest
The push refers to repository [docker.io/stasname/springboot-crud-app]
0fc91b052a5a: Pushed
c6b7fc72549c: Pushed
359efcd407b8: Pushed
1befe9768e55: Pushed
b1cba2e842ca: Pushed
f25d452629bd: Pushed
5a5ebddf856f: Pushed
90f93dcabaa5: Pushed
latest: digest: sha256:14c6e4bf009d94f76d5ae2f583c5c53c31c8c939dee166645d1a5f2a6a2e0acc size: 856
```

Repositories

All repositories within the **stasname** namespace.

Search by repository name | All content | Create a repository

Name	Last Pushed	Contains	Visibility	Scout
stasname/springboot-crud-app	in about 3 hours	IMAGE	Public	Inactive

1–1 of 1

7. Выполните следующие действия с целью изучить особенности сетевого взаимодействия:
   - Получить информацию о всех сетях, работающих на текущем хосте и подробности о каждом типе сети

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
74c3f955a1d5   bridge                        bridge    local
eabc8a17436e   host                          host      local
7905d7d158fa   none                          null      local
e50ac339160f   springboot_crud_app_app_net   bridge    local
3b6cdd6bfd02   springboot_crud_app_default   bridge    local
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "74c3f955a1d5bc64b6a44a008aa95c093b31fcab01dce81743821c21870ba69b",
        "Created": "2026-03-10T16:54:15.802327927Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.17.0.0/16",
                    "Gateway": "172.17.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.bridge.default_bridge": "true",
            "com.docker.network.bridge.enable_icc": "true",
            "com.docker.network.bridge.enable_ip_masquerade": "true",
            "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
            "com.docker.network.bridge.name": "docker0",
            "com.docker.network.driver.mtu": "1500"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.17.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect host
[
    {
        "Name": "host",
        "Id": "eabc8a17436e8dcbb88fb84350b2fc0cd77e22419755863e2b39cfe1ddca15d7",
        "Created": "2026-03-10T16:54:15.785813924Z",
        "Scope": "local",
        "Driver": "host",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": null
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {}
        }
    }
]
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect none
[
    {
        "Name": "none",
        "Id": "7905d7d158fab8edaa4c2b8bd6df36e6dedf1b7d6cc2e44c692e95a050aaba4c",
        "Created": "2026-03-10T16:54:15.762385735Z",
        "Scope": "local",
        "Driver": "null",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": null
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {},
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {}
        }
    }
]
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect springboot_crud_app_app_net
[
    {
        "Name": "springboot_crud_app_app_net",
        "Id": "e50ac339160f824fd90cadb21c0947fad747b2361e4eec74f4f587c7686fcc36",
        "Created": "2026-03-11T11:54:13.279177365Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.19.0.0/16",
                    "Gateway": "172.19.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {
            "com.docker.compose.config-hash": "6cf981445e660253296cfcd9a436c71dddcbc57e5893c7675ab99db2c3e2ad86",
            "com.docker.compose.network": "app_net",
            "com.docker.compose.project": "springboot_crud_app",
            "com.docker.compose.version": "5.0.2"
        },
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.19.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
```

- Создать свою собственную сеть bridge, проверить, создана ли она, запустить Docker-контейнер в созданной сети, вывести о ней всю информацию(включая IP-адрес контейнера), отключить сеть от контейнера

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network create my_bridge
294f4cd220c41cac5d32680aeb94d4eb96fbd9b4c2393067c2c7fc887fb1880a
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
74c3f955a1d5   bridge                        bridge    local
eabc8a17436e   host                          host      local
294f4cd220c4   my_bridge                     bridge    local
7905d7d158fa   none                          null      local
e50ac339160f   springboot_crud_app_app_net   bridge    local
3b6cdd6bfd02   springboot_crud_app_default   bridge    local
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app>
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect my_bridge
[
    {
        "Name": "my_bridge",
        "Id": "294f4cd220c41cac5d32680aeb94d4eb96fbd9b4c2393067c2c7fc887fb1880a",
        "Created": "2026-03-11T12:15:52.354464573Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.20.0.0/16",
                    "Gateway": "172.20.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.20.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker run -d --name test1 --network my_bridge alpine sleep 1000
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
589002ba0eae: Pull complete
9e595aac14e0: Download complete
caa817ad3aea: Download complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
ee4284393c6e2fd84101715e968c0098ad9f37d42ac6bbdb950c929aeaecec92
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker inspect test1
[
    {
        "Id": "ee4284393c6e2fd84101715e968c0098ad9f37d42ac6bbdb950c929aeaecec92",
        "Created": "2026-03-11T12:17:30.982561502Z",
        "Path": "sleep",
        "Args": [
            "1000"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 7111,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-03-11T12:17:31.22862993Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659",
        "ResolvConfPath": "/var/lib/docker/containers/ee4284393c6e2fd84101715e968c0098ad9f37d42ac6bbdb950c929aeaecec92/resolv.conf",
        "HostnamePath": "/var/lib/docker/containers/ee4284393c6e2fd84101715e968c0098ad9f37d42ac6bbdb950c929aeaecec92/hostname",
        "HostsPath": "/var/lib/docker/containers/ee4284393c6e2fd84101715e968c0098ad9f37d42ac6bbdb950c929aeaecec92/hosts",
        "LogPath": "/var/lib/docker/containers/ee4284393c6e2fd84101715e968c0098ad9f37d42ac6bbdb950c929aeaecec92/ee4284393c6e2fd84101715e968c0098ad9f37d42ac6bbdb950c929aeaecec92-json.log",
        "Name": "/test1",
        "RestartCount": 0,
        "Driver": "overlayfs",
        "Platform": "linux",
        "MountLabel": "",
        "ProcessLabel": "",
        "AppArmorProfile": "",
        "ExecIDs": null,
        "HostConfig": {
            "Binds": null,
            "ContainerIDFile": "",
            "LogConfig": {
                "Type": "json-file",
                "Config": {}
            },
            "NetworkMode": "my_bridge",
            "PortBindings": {},
            "RestartPolicy": {
                "Name": "no",
                "MaximumRetryCount": 0
            },
            "AutoRemove": false,
            "VolumeDriver": "",
            "VolumesFrom": null,
            "ConsoleSize": [
                48,
                204
            ],
            "CapAdd": null,
            "CapDrop": null,
            "CgroupnsMode": "private",
            "Dns": null,
            "DnsOptions": [],
            "DnsSearch": [],
            "ExtraHosts": null,
            "GroupAdd": null,
            "IpcMode": "private",
            "Cgroup": "",
            "Links": null,
            "OomScoreAdj": 0,
            "PidMode": "",
            "Privileged": false,
            "PublishAllPorts": false,
            "ReadonlyRootfs": false,
            "SecurityOpt": null,
            "UTSMode": "",
            "UsernsMode": "",
            "ShmSize": 67108864,
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network disconnect my_bridge test1
```

- Создать еще одну сеть bridge, вывести о ней всю информацию, запустить в ней три контейнера, подключиться к любому из контейнеров и пропинговать два других из оболочки контейнера,

убедиться, что между контейнерами происходит общение по IP-адресу

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network create my_bridge2
c4d855e1b8b9b996dd1ad9ba335626eab11618f098a313174fdcb4e811dc5f6b
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
74c3f955a1d5   bridge                        bridge    local
eabc8a17436e   host                          host      local
294f4cd220c4   my_bridge                     bridge    local
c4d855e1b8b9   my_bridge2                    bridge    local
7905d7d158fa   none                          null      local
e50ac339160f   springboot_crud_app_app_net   bridge    local
3b6cdd6bfd02   springboot_crud_app_default   bridge    local
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app>
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect my_bridge2
[
    {
        "Name": "my_bridge2",
        "Id": "c4d855e1b8b9b996dd1ad9ba335626eab11618f098a313174fdcb4e811dc5f6b",
        "Created": "2026-03-11T12:19:29.919348196Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.21.0.0/16",
                    "Gateway": "172.21.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.21.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker run -d --name c1 --network my_bridge2 alpine sleep 1000
e795c60a30fc6c812823e8e85a73158d3b44d178acfbde752b7e6001d8bf88eb
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker run -d --name c2 --network my_bridge2 alpine sleep 1000
b282f7679d6c975969870773041b5a6a0a35c46b299251bfe497d44e9c69eceb
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker run -d --name c3 --network my_bridge2 alpine sleep 1000
f95978427d2eca8c2cacaf70b5d8cf7f02be3717b951791501d4d5891b9321b4
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker exec -it c1 sh
/ # ping c2
PING c2 (172.21.0.3): 56 data bytes
64 bytes from 172.21.0.3: seq=0 ttl=64 time=1.235 ms
64 bytes from 172.21.0.3: seq=1 ttl=64 time=0.163 ms
64 bytes from 172.21.0.3: seq=2 ttl=64 time=0.126 ms
64 bytes from 172.21.0.3: seq=3 ttl=64 time=0.130 ms
64 bytes from 172.21.0.3: seq=4 ttl=64 time=0.166 ms
o64 bytes from 172.21.0.3: seq=5 ttl=64 time=0.140 ms
64 bytes from 172.21.0.3: seq=6 ttl=64 time=0.154 ms
64 bytes from 172.21.0.3: seq=7 ttl=64 time=0.147 ms
64 bytes from 172.21.0.3: seq=8 ttl=64 time=0.125 ms
q64 bytes from 172.21.0.3: seq=9 ttl=64 time=0.126 ms
64 bytes from 172.21.0.3: seq=10 ttl=64 time=0.138 ms
^C
--- c2 ping statistics ---
11 packets transmitted, 11 packets received, 0% packet loss
round-trip min/avg/max = 0.125/0.240/1.235 ms
```

```
round-trip min/avg/max = 0.125/0.240/1.233 ms
/ # ping c3
PING c3 (172.21.0.4): 56 data bytes
64 bytes from 172.21.0.4: seq=0 ttl=64 time=0.500 ms
64 bytes from 172.21.0.4: seq=1 ttl=64 time=0.146 ms
64 bytes from 172.21.0.4: seq=2 ttl=64 time=0.138 ms
64 bytes from 172.21.0.4: seq=3 ttl=64 time=0.167 ms
64 bytes from 172.21.0.4: seq=4 ttl=64 time=0.125 ms
64 bytes from 172.21.0.4: seq=5 ttl=64 time=0.146 ms
64 bytes from 172.21.0.4: seq=6 ttl=64 time=0.370 ms
^C
--- c3 ping statistics ---
7 packets transmitted, 7 packets received, 0% packet loss
round-trip min/avg/max = 0.125/0.227/0.500 ms
```

- Создать свою собственную сеть overlay, проверить, создана ли она, вывести о ней всю информацию

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker swarm init
Swarm initialized: current node (s992qacg8a5wp2cfm2cpee7t2) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-194s44hulfssyyirkt5mnwvy8yph5ybptdomzhv60j7hym34tf-1qrf2ecib8t3w7jcli1v2obtl 192.168.65.3:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app>
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network create -d overlay my_overlay
dxdwomzxuowcburt2sm60a4st
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
74c3f955a1d5   bridge                        bridge    local
bac757b86bd5   docker_gwbridge               bridge    local
eabc8a17436e   host                          host      local
hg89eorhj5yc   ingress                       overlay   swarm
294f4cd220c4   my_bridge                     bridge    local
c4d855e1b8b9   my_bridge2                    bridge    local
dxdwomzxuowc   my_overlay                    overlay   swarm
7905d7d158fa   none                          null      local
e50ac339160f   springboot_crud_app_app_net   bridge    local
3b6cdd6bfd02   springboot_crud_app_default   bridge    local
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect my_overlay
[
    {
        "Name": "my_overlay",
        "Id": "dxdwomzxuowcburt2sm60a4st",
        "Created": "2026-03-11T12:25:41.259685467Z",
        "Scope": "swarm",
        "Driver": "overlay",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "10.0.1.0/24",
                    "Gateway": "10.0.1.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.driver.overlay.vxlanid_list": "4097"
        },
        "Labels": null,
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "10.0.1.0/24": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 253
                    }
                }
            }
        }
    }
]
```

- Создать еще одну сеть overlay, проверить, создана ли она, вывести о ней всю информацию, удалить сеть

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network create -d overlay my_overlay2
zl1f42uv0vtsuxr1215fgf0mo
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network inspect my_overlay2
[
    {
        "Name": "my_overlay2",
        "Id": "zl1f42uv0vtsuxr1215fgf0mo",
        "Created": "2026-03-11T12:27:14.038056758Z",
        "Scope": "swarm",
        "Driver": "overlay",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "10.0.2.0/24",
                    "Gateway": "10.0.2.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.driver.overlay.vxlanid_list": "4098"
        },
        "Labels": null,
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "10.0.2.0/24": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 253
                    }
                }
            }
        }
    }
]
```

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
74c3f955a1d5   bridge                        bridge    local
bac757b86bd5   docker_gwbridge               bridge    local
eabc8a17436e   host                          host      local
hg89eorhj5yc   ingress                       overlay   swarm
294f4cd220c4   my_bridge                     bridge    local
c4d855e1b8b9   my_bridge2                    bridge    local
dxdwomzxuowc   my_overlay                    overlay   swarm
zl1f42uv0vts   my_overlay2                   overlay   swarm
7905d7d158fa   none                          null      local
e50ac339160f   springboot_crud_app_app_net   bridge    local
3b6cdd6bfd02   springboot_crud_app_default   bridge    local
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network rm my_overlay2
my_overlay2
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network ls
NETWORK ID     NAME                          DRIVER    SCOPE
74c3f955a1d5   bridge                        bridge    local
bac757b86bd5   docker_gwbridge               bridge    local
eabc8a17436e   host                          host      local
hg89eorhj5yc   ingress                       overlay   swarm
294f4cd220c4   my_bridge                     bridge    local
c4d855e1b8b9   my_bridge2                    bridge    local
dxdwomzxuowc   my_overlay                    overlay   swarm
7905d7d158fa   none                          null      local
e50ac339160f   springboot_crud_app_app_net   bridge    local
3b6cdd6bfd02   springboot_crud_app_default   bridge    local
```

- Попробовать создать сеть host, сохранить результат в отчет.

```
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app> docker network create -d host my_host
Error response from daemon: only one instance of "host" network is allowed
PS C:\Users\User\453502_SHAULOUSKI_26\IGI\LR2\springboot_crud_app>
```